Dollarization, Sovereign Default Risk, and Fiscal Rules*

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Abstract

This paper studies the optimal design of fiscal rules for a dollarized economy subject to sovereign default risk. When dollarization is modeled as a simple increase in income volatility (e.g., because the exchange rate cannot act as a shock-absorber), the dollarized economy prefers a tighter fiscal rule. This simple result is misleading: when the increase in income volatility is created by amplification through nominal rigidities, the optimal rule remains much closer to its floating exchange-rate benchmark. Different optimal rules result partly from higher conditional skewness in the economy with rigidities and partly from a higher marginal value of fiscal space in downturns, which disciplines the government. In a calibration to Ecuador, the optimal rule is a deficit limit of 1.7 percent of GDP and yields a welfare gain equivalent to 2.3 percent of permanent consumption relative to a no-rule benchmark. These results then recommend caution in applying reduced-form estimates to existing models.

JEL Classification E62, F34, F41, H63

Keywords dollarization, sovereign default risk, fiscal rules, sovereign debt, output volatility

*The views expressed herein are those of the authors and should not be attributed to the IMF or the World Bank, their Executive Boards, or their management.

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Introduction

Fiscal rules have become widely adopted tools of macroeconomic governance in recent decades. Numerical constraints on government budgetary aggregates such as debt, deficits, or expenditures have proliferated across more than 100 countries since 1985, according to the IMF Fiscal Rules Database (Alonso-Albarran et al., 2025). An active literature studies the effect and design of fiscal rules by leveraging sovereign debt and default models, in which the presence of long-term debt creates time inconsistency and enables gains from committing to rules that constrain debt issuances ex-post. This literature finds that tailoring the fiscal rule to country characteristics such as output volatility determines the size and sign of welfare gains delivered by fiscal rules. For dollarized economies, or very credible exchange-rate pegs, which are typically more volatile than comparable economies with more flexible exchange-rate regimes (Edwards and Magendzo, 2006; Korab et al., 2023), the careful design of fiscal rules is particularly important.

In this paper we study the problem of designing a fiscal rule for a dollarized economy. We analyze the welfare gains and associated tradeoffs of different fiscal rules in the context of a sovereign debt and default model tailored to the case of dollarization. We introduce dollarization in this model in two possible ways: as a reduced-form increase in the volatility of output, and explicitly via a traded-nontraded structure and nominal wage rigidities (similar to the pegged exchange rate case of Bianchi and Sosa-Padilla, 2024). In each case, we consider different designs for the fiscal rule, including limits to the debt level, to the deficit, and debt-brake rules which limit deficits only when debt exceeds a certain threshold. We compute the optimal parameters for each rule and compare their welfare gains relative to a no-rule benchmark.

When dollarization is modeled as a simple increase in output volality, the economy chooses a tighter fiscal rule (both in the debt limit and deficit limit cases) and attains marginally lower welfare gains from adopting the optimal rule than a less volatile economy. Moreover, using the debt limit that is optimal for a different economy (but too tight for the current one) can easily lead to welfare losses relative to the no-rule case. However, when dollarization is modeled explicitly as a tightening of nominal rigidities (which do increase output volatility endogenously), the optimal fiscal rule becomes insensitive to the strength of nominal rigidities. This structural approach also implies larger welfare gains of fiscal rules for the dollarized economy relative to the reduced-form volatile but freely-floating one.

We also consider the possibility that dollarization also affects trend growth. Higher trend growth mechanically improves $r - g$ and debt sustainability and therefore allows the planner to choose a looser fiscal rule. More interestingly, the gains from adopting a fiscal rule are increasing in the speed of trend growth.

Discussion of the Literature To be written.

Layout The rest of paper is structured as follows. Section 2 describes the model. Section 3 describes the calibration strategy and results. Section 4 presents the quantitative results regarding fiscal rules. Finally, Section 5 concludes.

2. Model

We consider a TNT setup with traded and nontraded goods. The small open economy receives an endowment of traded goods and produces nontradables with labor, subject to wage rigidities. The government issues debt in international markets. The government acts in the best interest of domestic households but lacks commitment to future actions, including borrowing and debt repayment choices. We focus on the case of a fixed exchange rate to represent dollarization and compare it with different cases with exchange-rate flexibility.

Resources The economy receives an endowment stream of tradable goods following a stochastic process with trend and cycle components:

$$Y_{T,t} = \exp(z_t)\Gamma_t \tag{1}$$

with

$$\begin{aligned} z_t &= \rho_z z_{t-1} + \sigma_z \epsilon_t^z \\ \log(\Gamma_t) &= \log(\Gamma_{t-1}) + \mu \end{aligned}$$

where $\epsilon_t^z \overset{iid}{\sim} \mathcal{N}(0,1)$.

Domestic firms produce nontraded goods with labor according to

$$Y_N = \Gamma f(h) = \Gamma h^{\alpha}$$

World prices The price of tradable goods p_T is determined in global markets with fluctuations that are shocks from the perspective of the small open economy:

$$\log(p_t^T) = \rho_p \log(p_{t-1}^T) + \sigma_p \epsilon_t^p$$

where $\epsilon_t^p \overset{iid}{\sim} \mathcal{N}(0,\sigma_p)$.

Households Households consume a CES aggregate of tradable and nontradable goods, with elasticity of substitution $1/(1+\eta)$

$$C = \mathscr{C}(C_N, C_T) = \left[\varpi_N C_N^{-\eta} + \varpi_T C_T^{-\eta}\right]^{-\frac{1}{\eta}}$$

where $(\varpi_N, \varpi_T) > 0, \varpi_N + \varpi_T = 1$, and $\eta > -1$. As a consequence, the consumer price index is

$$P_c = \left[\varpi_N^{\frac{1}{1+\eta}} p_N^{\frac{\eta}{1+\eta}} + \varpi_T^{\frac{1}{1+\eta}} p_T^{\frac{\eta}{1+\eta}}\right]^{\frac{1+\eta}{\eta}}$$

Another consequence of CES preferences is the first-order condition for the composition of consumption given the relative price of nontradables:

$$\frac{p_N}{p_T} = \frac{\varpi_N}{\varpi_T}\left(\frac{C_T}{C_N}\right)^{1+\eta} \tag{2}$$

Finally, households face a budget constraint equating the value of expenditures on consumption to labor income, profits from the firms, and lump-sum taxes or transfers from the government. As in many models of sovereign debt, rich tax instruments enable the government to make consumption choices directly.

Labor markets We assume that wages cannot fall below some level $\Gamma\bar{w}$, while households supply labor inelastically up to $\bar{h} = 1$. Whenever the constraint on wages binds, the production of nontradables is constrained at $\bar{y}_N = \Gamma f(\bar{h})$. Combining the equilibrium relative price of nontradables with the labor demand equation $w = p_N f'(h)$, we obtain the familiar relation between employment in the nontradable sector and consumption of tradables

$$h \leq \mathscr{H}(C_T/\Gamma, \bar{w}, p_T) = \left(p_T \frac{\varpi_N}{\varpi_T}\frac{\alpha}{\bar{w}}\right)^{\frac{1}{1+\alpha\eta}} \left(\frac{C_T}{\Gamma}\right)^{\frac{1+\eta}{1+\alpha\eta}}$$

Assets The government borrows from international lenders in the form of a defaultable bond which promises to pay a non-contingent stream of geometrically-decaying coupons. A bond issued in period t pays $(1-\delta)^{s-1}\kappa$ units of the tradable good in period $t+s$, effectively making a one-period-old bond a perfect substitute of $(1-\delta)$ units of newly-issued debt. The coupon rate $\kappa = r + \delta$, where r is the international risk-free rate, is chosen so that the price of a bond that is expected to never default is $q^* = 1$. This normalization ensures that one unit of the bond corresponds to a face value of one unit of the tradable good.

Upon default, the government loses access to international capital markets and faces a loss of output. There is uncertainty about the length of the exclusion period. Most of the literature assumes that the resumption of debt service payments, the recovery of market access, and the dissipation of any output costs all happen at the same time. We keep this assumption.

Government The government is benevolent and makes choices on a sequential basis to maximize the utility of a representative household with preferences given by

$$V_t \;=\; \mathbb{E}_t\left[\sum_{s=0}^{\infty}\beta^s\Big(u(\mathscr{C}(C^N_{t+s}, C^T_{t+s})) \;+\; \Gamma^{1-\gamma}_{t+s}\,\epsilon_{t+s}\Big)\right] \tag{3}$$

where $\mathbb{E}\left[\cdot\right]$ denotes the expectation operator, (C^N_t, C^T_t) represents the household's consumption vector, β is a discount factor, and ϵ is a preference shock for default or repayment, scaled by $\Gamma^{1-\gamma}$ to ensure that its importance for the government's decisions remains commensurate to relevant quantities in the face of a non-stationary endowment process (see below).

While the government is not in default, it chooses whether to repay the debt and attains a value

$$V(B, z, \Gamma, p_T) \;=\; \mathbb{E}\left[\max\left\{V_R(B, z, \Gamma, p_T) + \Gamma^{1-\gamma}\epsilon_R,\; V_D(z, \Gamma, p_T) + \Gamma^{1-\gamma}\epsilon_D\right\}\right], \tag{4}$$

where the ϵ's follow a Type 1 Extreme Value distribution. This yields closed forms for the value function in the normalized formulation below.

Debt issuances If the government chooses to repay the debt, it can access capital markets and issue new debt B', so that

$$\begin{aligned}
V_R(B, z, \Gamma, p_T) \;&=\; \max_{B'}\; u(\mathscr{C}(C_N, C_T)) + \beta\mathbb{E}\left[V(B', z', \Gamma e^{\mu}, p_T') \mid z, p_t\right] \\
\text{subject to}\quad & p_T C_T + \kappa B = p_T Y_T(z, \Gamma) \;+\; q(B', z, \Gamma, p_T)\left(B' - (1-\delta)B\right) \\
& C_N = \Gamma f(h) \\
& h = \min\left\{1, \mathscr{H}(C_T/\Gamma, \bar{w}, p_T)\right\}
\end{aligned} \tag{5}$$

Default decision If the government chooses to default, it loses access to international capital markets, which it then recovers with a constant hazard ψ. While it is excluded, the borrowing economy suffers a loss of output $h(z)$. We therefore maintain the usual assumption that the resumption of payments and the renewed access to international markets happen simultaneously.

We assume that all debt is destroyed after a default for tractability and to more readily compare with other results in the literature. The value of default is

$$\begin{aligned}
V_D(z, \Gamma, p_T) &= u(\mathscr{C}(C_N, C_T)) + \beta\mathbb{E}_{z, p_T}\left[\psi V(0, z', \Gamma e^{\mu}, p_T') + (1-\psi)V_D(z', \Gamma e^{\mu}, p_T')\right] \\
\text{subject to}\quad & C_T = Y_T(z, \Gamma)(1 - h(z)) \\
& C_N = \Gamma f(h) \\
& h = \min\left\{1, \mathscr{H}(C_T/\Gamma, \bar{w}, p_T)\right\}
\end{aligned} \tag{6}$$

The output loss function is defined as $h(z) \;=\; \max\{0, d_1 \,+\, d_2 \exp(z)\}$, where d_1 and d_2 are parameters to be calibrated.

Lenders Bonds issued by the government are purchased by deep-pocketed, risk-neutral foreign investors who equate the expected return of the debt to their cost of funds r, yielding a debt price

$$q(B', z, \Gamma, p_T) = \frac{1}{1+r}\,\mathbb{E}\left[\big(1 - \mathscr{P}(B', z', \Gamma e^{\mu}, p_T')\big)\big(\kappa + (1-\delta) q(B'', z', \Gamma e^{\mu}, p_T')\big) \mid z, p_T\right] \tag{7}$$

where B'' and $\mathscr{P}$ denote the government's issuance and default policy expected by lenders. We implicitly assume these are private investors.

2.1 Equilibrium

We define an equilibrium of the economy in its original non-stationary form before proceeding to normalizing the model to compute a solution.

Definition. *A Markov-Perfect equilibrium consists of a set of value functions V, V_R, and V_D, a default probability $\mathscr{P}$, a borrowing rule B', and a bond price function q, such that*

1. *Given the price of debt, the value functions satisfy functional equations (4)–(6).*
2. *Given the default probability and borrowing rules, the price of debt satisfies the creditors' Euler equation (7).*
3. *Given the price of debt, the default probability and borrowing rules correspond to the policy functions solving (4) and (5).*

2.2 Normalization

We detrend variables by Γ_t and denote normalized values with lowercase. For example, $y_{T,t} = Y_{T,t}/\Gamma_t = \exp(z_t)$. We also specialize the utility function to a CRRA form, and obtain

$$u(C) = \frac{C^{1-\gamma}}{1-\gamma} = \Gamma^{1-\gamma}\,\frac{(C/\Gamma)^{1-\gamma}}{1-\gamma} = \Gamma^{1-\gamma}\,u(c)$$

which implies a normalization constant $\Gamma_t^{1-\gamma}$ for the value functions.

Normalizing variables by Γ leads to a formulation with state variables (b, z, p_T). At the beginning of a period, the government faces the choice of default and repayment (4), where we guess-and-verify that $V(B, z, \Gamma, p_T) = \Gamma^{1-\gamma} v(b, z, p_T)$ and multiply both sides by $\Gamma^{\gamma-1}$ to obtain

$$v(b, z, p_T) = \mathbb{E}\big[\max\{v_R(b, z, p_T) + \epsilon_R,\ v_D(z, p_T) + \epsilon_D\}\big] \tag{8}$$

The distributional assumption for the ϵ's implies closed forms for the value function $v(b, z, p_T)$ and the (ex-post) default probability $\mathscr{P}(b, z, p_T)$:

$$
\begin{aligned}
v(b,z,p_T) &= \chi \log\big(\exp(v_D(z,p_T)/\chi) + \exp(v_R(b,z,p_T)/\chi)\big), \\
\mathscr{P}(b,z,p_T) &= \frac{\exp(v_D(z,p_T)/\chi)}{\exp(v_D(z,p_T)/\chi) + \exp(v_R(b,z,p_T)/\chi)}
\end{aligned} \tag{9}
$$

For the values of repayment and default, the normalization constant manifests as an 'effective' discount factor $\tilde{\beta} = \beta \exp(\mu)^{1-\gamma}$ which, as is standard in non-stationary environments, incorporates the growth rate of the economy. As a result, trend growth and impatience impact borrowing decisions in the same way. The normalization also drives a wedge between the control variable B'/Γ which appears in the budget constraint and the following period's state B'/Γ' which appears in the continuation value function. To reserve the symbol b' for the second meaning, we denote the current period's issuance B'/Γ as x. Clearly, it is equivalent to choose b' or x as long as the relationship between them, $b' = x/e^{\mu}$, is imposed where appropriate. The value function for repayment is

$$
\begin{aligned}
v_R(b,z,p_T) &= \max_x u(\mathscr{C}(c_N,c_T)) + \beta(e^{\mu})^{1-\gamma}\mathbb{E}\Big[v(x/e^{\mu},z',p_T')\Big| z,p_T\Big] \\
\text{subject to} \quad & p_T c_T + \kappa b = p_T y_T(z) + q(x,z,p_T)(x-(1-\delta)b); \\
& c_N = f(h); \\
& h = \min\{1, \mathscr{H}(c_T,\bar{w},p_T)\}
\end{aligned} \tag{10}
$$

The value of default incorporates the same effective discounting:

$$
\begin{aligned}
v_D(z,p_T) \; &= u(\mathscr{C}(c_N,c_T)) + \beta(e^{\mu})^{1-\gamma}\mathbb{E}\left[\psi v(0,z',p_T') + (1-\psi)v_D(z',p_T') \mid z,p_T\right] \\
\text{where} \quad c_T &= y_T(z)(1-h(z)) \\
c_N &= f(h) \\
h &= \min\{1, \mathscr{H}(c_T,\bar{w},p_T)\}
\end{aligned} \tag{11}
$$

The price of debt (7) is relatively unaffected by the normalization but features the distinction between $x = B'/\Gamma$, the debt chosen in the current period, and $b' = B'/\Gamma'$, the debt outstanding at the beginning of the following period

$$
q(x,z,p_T) = \frac{1}{1+r}\mathbb{E}\Big[\big(1-\mathscr{P}(x/e^{\mu},z',p_T')\big)\big(\kappa + (1-\delta)q(x',z',p_T')\big)\Big| z,p_T\Big] \tag{12}
$$

2.3 *Fiscal Rules*

We consider three types of fiscal rules: a debt limit rule, a deficit limit rule, and a debt brake rule. The first two are static rules that impose ceilings on the level of debt and the fiscal deficit, respectively. The third is a dynamic rule that combines both instruments: once the debt threshold is breached, a deficit limit is triggered.

To compute fiscal-rule statistics, we compute debt and deficit over total (annual) GDP as

$$\begin{aligned} \textit{Debt-to-GDP} &= \frac{b}{4\,(p_T y_T + p_N y_N)} \\ \textit{Deficit-to-GDP} &= \frac{q \cdot (b' - (1-\delta)b)}{4\,(p_T y_T + p_N y_N)} = \frac{p_T y_T - p_T c_T - \kappa b}{4\,(p_T y_T + p_N y_N)} \end{aligned} \tag{13}$$

where we emphasize that the government's budget constraint allows computing the (overall) deficit as net borrowing or as the difference between revenues and expenditures (including interest).

In the model with a debt limit rule (with limit $\bar{d}$), the government is constrained to choosing under the constraint that *Debt-to-GDP* $\leq \bar{d}$. Under a deficit rule with limit $\bar{\delta}$, the government is constrained to choosing under the constraint that *Deficit-to-GDP* $\leq \bar{\delta}$. With a debt-brake with limits $(\bar{d}, \bar{\delta})$, the government is subject to *Deficit-to-GDP* $\leq \bar{\delta}$ every time its initial debt exceeds the debt limit $\bar{d}$.[1]

In Section 4, we compute the optimal parameters for each rule by comparing their welfare gains relative to the no-rule benchmark. We then study the sensitivity of these results to output volatility by varying σ_z and $\bar{w}$.

3. Calibration

We calibrate the model to Ecuador, a dollarized economy, by matching statistics of a 40-quarter pre-default sample to data for the period 2010q1–2019q4. We use this period, in which a sovereign default did not occur, to estimate the parameters of the exogenous processes and compute calibration targets. Table 1 summarizes our parametrization. Table 2 reports the fit of the model.

[1]The presence of endogenous objects (p_N, y_N) in the denominator of the debt ratio means that some choices for debt issuance may increase GDP and hence marginally relax the constraint.

Table 1: Baseline Parameter Values

Externally chosen		
	Parameter	**Value**
Sovereign's risk aversion	γ	2
Preference shock scale parameter	χ	0.0005
Preference shock scale parameter	χ_b	0.0005
Risk-free interest rate	r	0.01
Duration of debt	δ	0.05
Reentry probability	ψ	0.0385
Tradable output autocorrelation coefficient	ρ_z	0.6651
Standard deviation of z_t	σ_z	0.0236
Tradable price autocorrelation coefficient	ρ_p	0.9863
Standard deviation of p_T	σ_p	0.0195
Tradable output trend growth	μ	0.0052
Share of nontradables in consumption	ϖ_N	0.4183
Share of tradables in consumption	ϖ_T	0.5817
Elasticity of substitution between c_N and c_T	η	0.74
Labor share in production	α	0.67
Internally calibrated		
	Parameter	**Value**
Sovereign's discount factor	β	0.9084
Default cost: linear	d_1	-0.3146
Default cost: quadratic	d_2	0.3768
Minimum wage	$\bar{w}$	0.4906

Table 2: Model Fit

Description	**Data**	**Model**
Average external debt-to-GDP ratio, %	21.8	21.7
Average spread, bps	774	656
Std. Dev. spread, bps	220	268
Unemployment rate, %	4.5	4.42

Note: Simulation moments are based on 40-quarter samples before a default on market debt, conditional on no default occurring within those 40 quarters or two years prior. Data moments are computed for Ecuador between 2010q1 and 2019q4. Model moments are computed over repayment periods only.

4. Quantitative Results

We summarize the optimal fiscal rules and their associated welfare gains in Table 3. The optimal debt limit is effectively infinite, which implies that there is not a binding debt limit rule that yields welfare gains relative to the no-rule baseline. The optimal deficit limit is 1.7 percent of GDP and yields a welfare gain of 2.3 percent of permanent consumption, while the optimal debt brake rule is an unconditional deficit limit of 1.7 percent of GDP.

Table 3: Optimal Fiscal Rules Setting and Moments

	Baseline	**Debt Limit** $\bar{d} = +\infty$	**Deficit Limit** $\bar{\delta} = 1.7\%$	**Debt Brake** $(\bar{d}, \bar{\delta}) = (0, 1.7\%)$
Debt-to-GDP (%)	19.6	19.6	27.1	27.1
Spread (bps)	632	632	58	58
SD. Spread (bps)	327	327	62	62
Default Freq. (%)	8.5	8.5	0.5	0.5
Unemp. (%)	3.96	3.96	4.91	4.91
Welfare Gain (%)	–	–	2.3	2.3

Note: Each type of rule is evaluated at its optimal parametrization. Welfare gains are relative to a no-rule benchmark and expressed as percent of permanent consumption. Statistics are based on simulations of the ergodic distribution of each model, conditional on repayment.

Since our model is calibrated to a dollarized economy, which is typically more volatile than a comparable non-dollarized economy (Edwards and Magendzo, 2006; Korab et al., 2023), it is useful to assess the sensitivity of our results to output volatility. We therefore vary σ_z and $\bar{w}$ and recompute the welfare gains under each fiscal rule.

As Figure 1 shows, the optimal rules become tighter as output volatility increases. However, the optimal (looser) rule yields marginally higher welfare gains when output volatility is low. With lower underlying volatility, the government is able to achieve a reduction in default risk with a looser rule that does not overly restrict deficits or consumption smoothing. At higher volatility, this tradeoff shifts as stabilizing spreads requires tighter limits.

However, as Figure 2 shows, the optimal fiscal rules are less sensitive to the friction introduced by the dollarization captured by the minimum wage $\bar{w}$. Having said that, it can be seen that the welfare gains associated with the implementation of fiscal rules are bigger in the case of a dollarized economy ($\bar{w} > 0$) than a flexible economy ($\bar{w} = 0$).

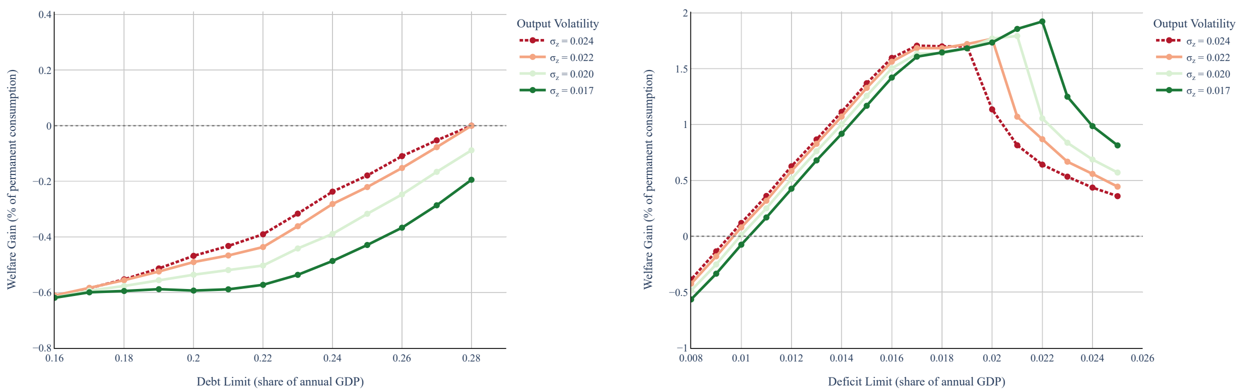
Finally, taking advantage of all characteristics of our model, we also assess the sensitivity of our results to the output trend growth μ and find, as Figure 3 shows, the optimal fiscal rules become looser as the output trend growth increases.

Taken together, these results highlight the importance of calibrating fiscal rules according to underlying macroeconomic fundamentals.

Figure 1: Optimal Fiscal Rules - Sensitivity to Output Volatility (σ_z) with $\bar{w} = 0$

(a) Debt Limit Rule

(b) Deficit Limit Rule



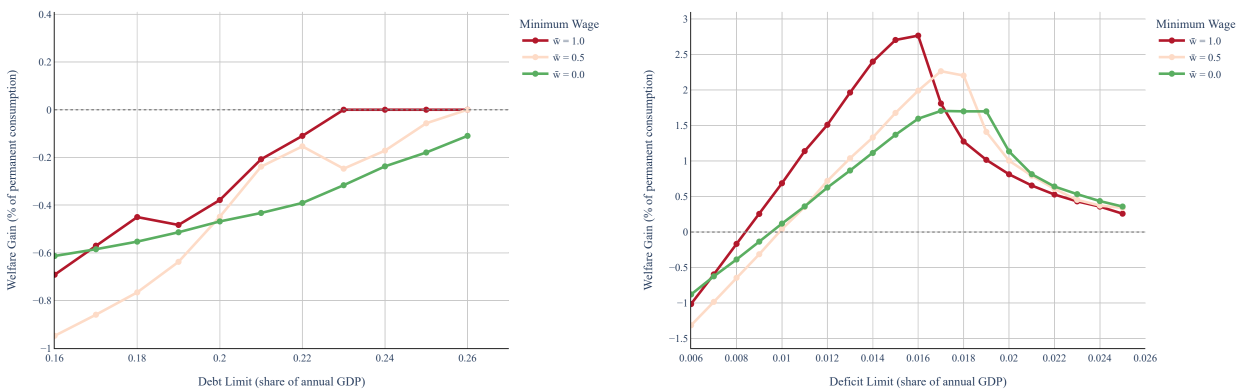
Note: The $\sigma_z = 0.024$ line corresponds to the calibrated dollarized economy.

Welfare gains are relative to a no-rule benchmark.

Figure 2: Optimal Fiscal Rules - Sensitivity to Minimum Wage ($\bar{w}$)

(a) Debt Limit Rule

(b) Deficit Limit Rule



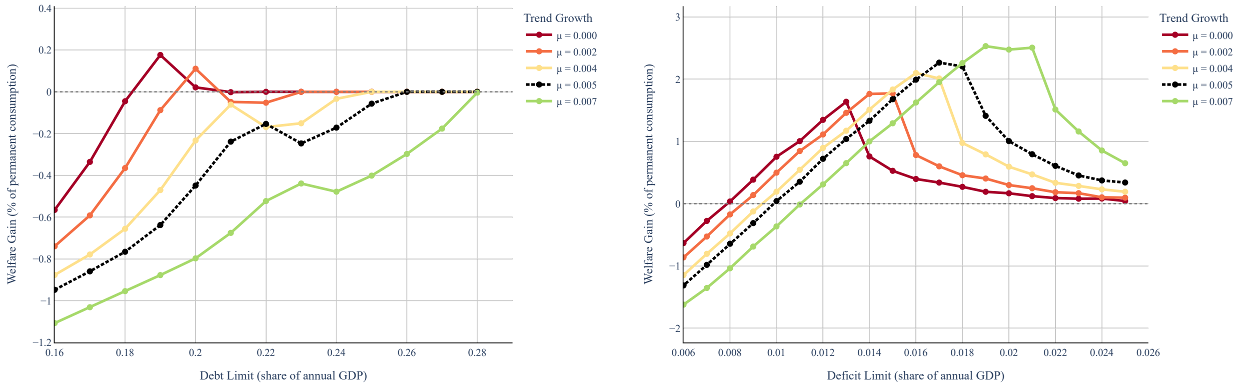
Note: The $\bar{w} = 0.5$ line corresponds to the calibrated dollarized economy.

Welfare gains are relative to a no-rule benchmark.

Figure 3: Optimal Fiscal Rules - Sensitivity to Output Trend Growth (μ)

(a) Debt Limit Rule

(b) Deficit Limit Rule



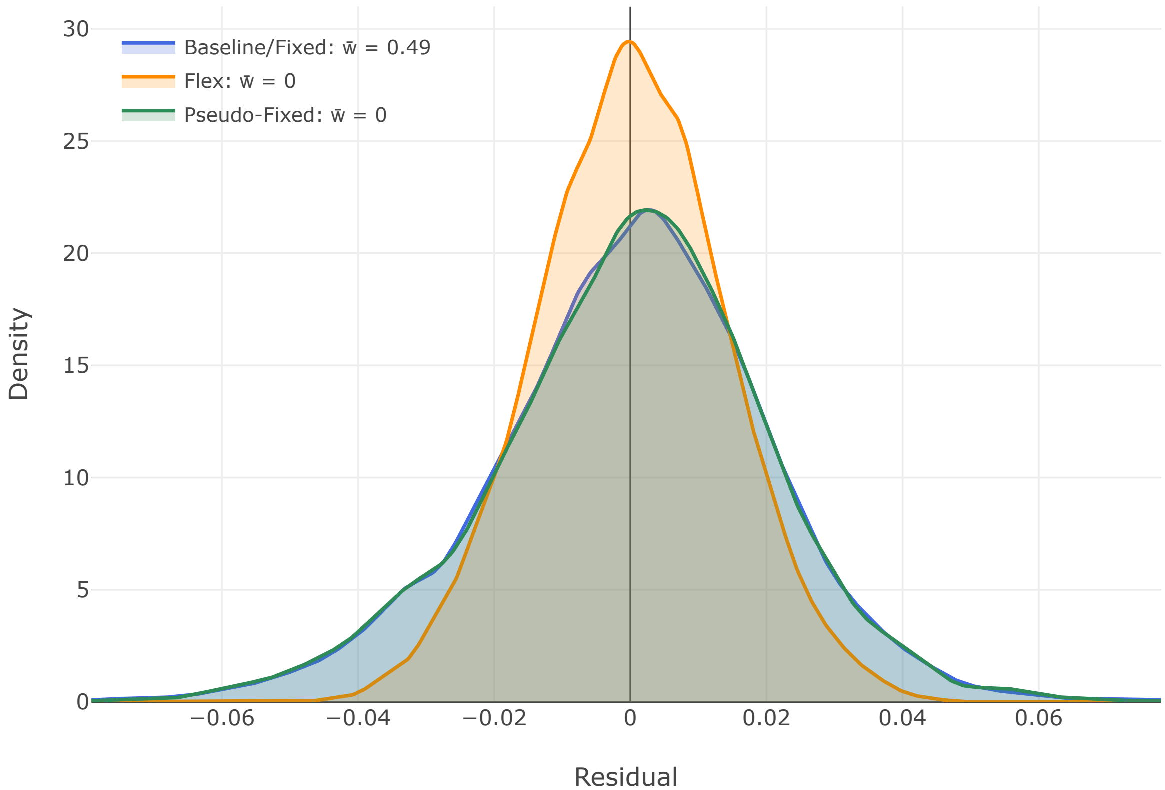
Note: The $\mu = 0.005$ line corresponds to the calibrated dollarized economy.

Welfare gains are relative to a no-rule benchmark.

4.1 A pseudo-fixed model

A flex model ($\bar{w} = 0$) that implies the same stochastic process for real GDP as the fixed model.

Figure 4: Distributions of AR(1) Residuals for ln(Real GDP)



Note: The figure reports kernel-density estimates of the residuals u_t from model-specific OLS regressions $\log Y_t = a + \rho \log Y_{t-1} + u_t$, where $Y_t = (p_{T,t} y_{T,t} + p_{N,t} y_{N,t})/P_{c,t}$. Each model is simulated for 8,000 retained quarterly periods after a 1,000-period burn-in, using the same seed and primitive random draws. "Baseline" is the fixed economy with wage rigidity; "Flex" sets $\bar{w} = 0$ while retaining Normal tradable-output innovations; and "Pseudo-Fixed" sets $\bar{w} = 0$ but uses a generalized Tauchen process based on the kernel-smoothed distribution of baseline residuals, with its innovation scale calibrated to match baseline residual volatility.

Table 4: Optimal Fiscal Rules for the Pseudo-Fixed Model

	Baseline	**Debt Limit** $\bar{d} = 16\%$	**Deficit Limit** $\bar{\delta} = 1.1\%$	**Debt Brake** $(\bar{d}, \bar{\delta}) = (0, 1.1\%)$
Debt-to-GDP (%)	14.4	14.2	16.3	16.3
Spread (bps)	1439	499	0.25	0.25
SD. Spread (bps)	401	103	0.82	0.82
Default Freq. (%)	8.9	5.2	0.0	0.0
Unemp. (%)	–	–	–	–
Welfare Gain (%)	–	0.4	1.5	1.5

Note: Each type of rule is evaluated at its optimal parametrization. Welfare gains are relative to a no-rule benchmark and expressed as percent of permanent consumption. Statistics are based on simulations of the ergodic distribution of each model, conditional on repayment.

[discussion]

5. Concluding Remarks

This paper studies the design of fiscal rules in a dollarized economy subject to sovereign default risk. In our calibration to Ecuador, the tradeoff between reducing default risk and restricting debt issuance is resolved by deficit limits, which are preferred to debt limits. The optimal deficit limit applies at all levels of debt; this is, the optimal debt brake rule is an unconditional deficit rule.

The results also show that the source of dollarization-related volatility matters for policy design. When dollarization is represented as higher exogenous output volatility, the optimal fiscal rule becomes tighter as volatility increases. However, when dollarization is modeled explicitly through a tradable-nontradable structure with nominal rigidities and a fully credible permanently pegged exchange rate, the desired tightness of the rule is much less sensitive to the degree of rigidity, even though the welfare gains from adopting a rule may be larger for a dollarized economy. Finally, higher output trend growth relaxes the optimal fiscal constraint by improving debt sustainability. Taken together, these findings suggest that fiscal rules should be carefully calibrated to the underlying macroeconomic characteristics.

A. Estimation of Exogenous Processes

Table 5: AR(1) for Tradable Output Process (z)

N = 40	Coefficient	SE	p-value	95% CI	
ρ_z	0.665	0.121	0.000	0.427	0.903
σ_z	0.024	0.002	0.000	0.018	0.029

Note: We estimate an AR(1) process over the linear detrended ln(Real Tradable Output Index) of Ecuador during 2010q1-2019q4.

Table 6: AR(1) for Tradable Price Process (p_T)

N = 120	Coefficient	SE	p-value	95% CI	
ρ_p	0.995	0.009	0.000	0.977	1.013
σ_p	0.011	0.001	0.000	0.010	0.013

Note: We estimate an AR(1) process over the ln(Real Effective Exchange Rate) of Ecuador during 2010m1-2019m12. The reported values were estimated in monthly frequency and converted to quarterly frequency using the formulas $\rho_q = \rho_m^3$ and $\sigma_q = \sigma_m\sqrt{1 + \rho_m^2 + \rho_m^4}$.

B. Additional Tables and Figures

Table 7: Debt Limit Rule Moments

limit	**welfare**	**dfreq**	**debtGDP**	**spread**	**stdspr**	**unemp**
0	-2.21	0	0	0.01	0	4.02
0.01	-2.12	0	0.70	0.01	0	4.03
0.02	-2.02	0	1.68	0.02	0	4.08
0.03	-1.94	0	2.66	0.01	0	4.12
0.04	-1.86	0	3.64	0.02	0	4.16
0.05	-1.79	0	4.62	0.02	0	4.21
0.06	-1.72	0	5.58	0.02	0	4.25
0.07	-1.65	0	6.55	0.03	0	4.29
0.08	-1.58	0	7.51	0.03	0.00	4.34
0.09	-1.51	0	8.46	0.03	0.00	4.38
0.10	-1.44	0	9.41	0.04	0.05	4.42
0.11	-1.37	0	10.3	0.07	0.22	4.47
0.12	-1.29	0	11.3	0.23	0.66	4.51
0.13	-1.21	0	12.2	0.80	1.64	4.55
0.14	-1.12	0	13.1	2.53	3.77	4.60
0.15	-1.04	0.08	14.0	7.75	7.71	4.64
0.16	-0.95	0.16	14.9	18.7	13.1	4.70
0.17	-0.86	0.41	15.8	38.6	23.9	4.72
0.18	-0.77	0.68	16.7	68.9	36.8	4.59
0.19	-0.64	1.41	17.4	108	55.4	4.52
0.20	-0.45	2.46	18.2	163	82.5	4.40
0.21	-0.24	4.02	18.7	256	125	4.12
0.22	-0.15	6.79	19.0	430	239	4.03
0.23	-0.25	9.00	18.9	626	317	3.83
0.24	-0.17	9.49	19.1	664	359	3.74
0.25	-0.06	9.29	19.4	642	344	3.90

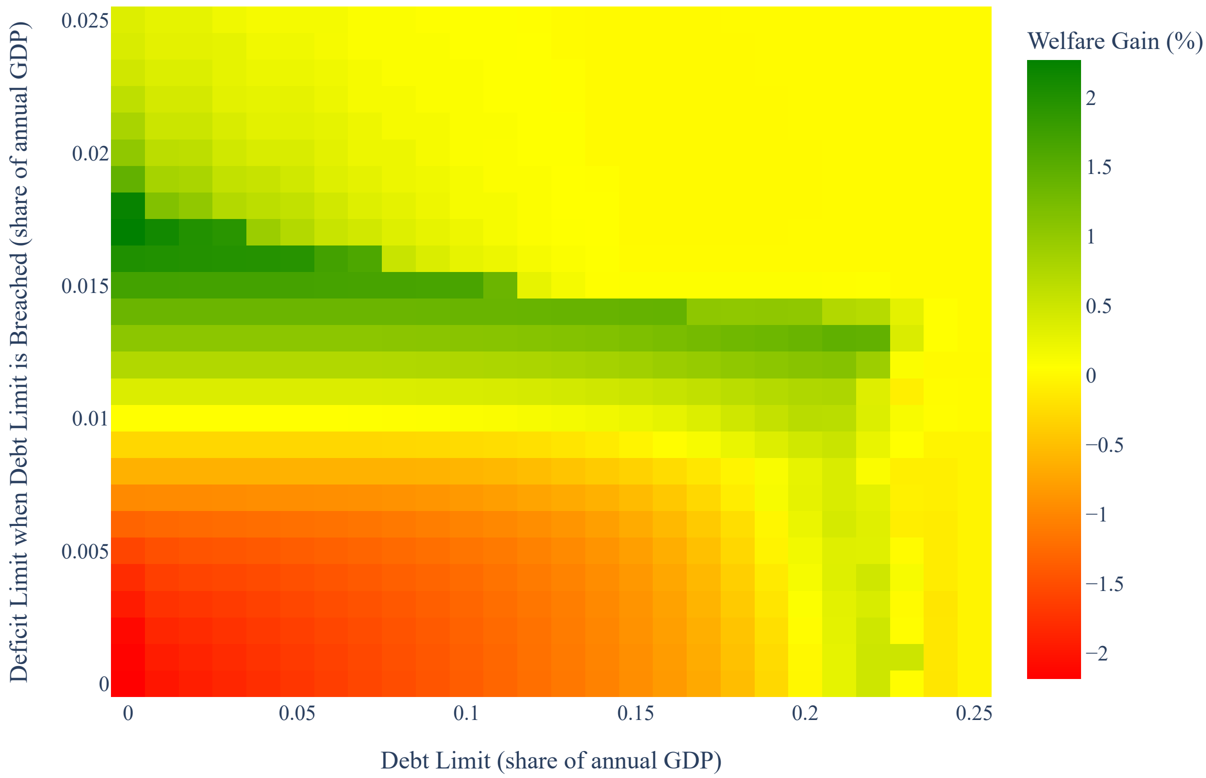
Note: There is not a debt limit that yields welfare gains relative to the no-rule benchmark. Statistics are based on simulations of the ergodic distribution of each model, conditional on repayment.

Table 8: Deficit Limit Rule Moments

limit	welfare	dfreq	debtGDP	spread	stdspr	unemp
0	-2.19	0	0	0.03	0	4.02
0.001	-2.16	0	0	0.03	0	4.02
0.002	-2.11	0	0.05	0.00	0	4.03
0.003	-1.96	0	1.46	0.00	0	4.15
0.004	-1.80	0	3.29	0.01	0	4.21
0.005	-1.59	0	5.13	0.01	0	4.27
0.006	-1.31	0	6.96	0.01	0	4.32
0.007	-0.98	0	8.79	0.01	0.00	4.37
0.008	-0.64	0	10.6	0.00	0	4.43
0.009	-0.31	0	12.5	0.00	0.00	4.47
0.010	0.04	0	14.3	0.00	0.00	4.52
0.011	0.35	0	16.1	0.00	0	4.58
0.012	0.72	0	18.0	0.00	0.00	4.63
0.013	1.04	0	19.8	0.00	0.00	4.69
0.014	1.33	0	21.7	0.02	0.07	4.74
0.015	1.68	0	23.5	0.20	0.74	4.80
0.016	1.99	0	25.4	6.46	23.2	4.87
0.017	2.27	0.49	27.1	57.7	62.5	4.91
0.018	2.20	2.76	25.9	247	148	4.83
0.019	1.41	5.05	22.4	405	227	4.35
0.020	1.00	6.40	21.1	459	262	4.16
0.021	0.79	7.19	20.6	490	273	3.91
0.022	0.61	7.72	20.2	518	311	3.87
0.023	0.45	7.68	20.1	542	305	3.98
0.024	0.37	7.93	19.9	555	308	3.84
0.025	0.34	8.55	19.8	560	346	3.74

Note: Maximum welfare gain of 2.3% with a deficit-to-GDP limit of 1.7%. Welfare gains are relative to a no-rule benchmark and expressed as percent of permanent consumption. Statistics are based on simulations of the ergodic distribution of each model, conditional on repayment.

Figure 5: Debt Brake Rule - Welfare Gains



Note: Maximum welfare gain of 2.3% with a debt-to-GDP limit of 0% and a deficit-to-GDP limit of 1.7%. Welfare gains are relative to a no-rule benchmark and expressed as percent of permanent consumption.

References

Alonso-Albarran, V., C. Arroyo, O. Aydin, H. Davoodi, W. Lam, V. Balasundharam, G. Hegab, A. Nguyen, N. Salazar, G. Sher, A. Solovyeva, and N. Tchelishvili (2025): "Fiscal Rules and Fiscal Councils," *IMF Working Papers*, 2025, 1.

Bianchi, J. and C. Sosa-Padilla (2024): "Reserve Accumulation, Macroeconomic Stabilization, and Sovereign Risk," *The Review of Economic Studies*, 91, 2053–2103.

Edwards, S. and I. I. Magendzo (2006): "Strict Dollarization and Economic Performance: An Empirical Investigation," *Journal of Money, Credit and Banking*, 269–282.

Korab, P., J. Fidrmuc, and S. Dibooglu (2023): "Growth and inflation tradeoffs of dollarization: Meta-analysis evidence," *Journal of International Money and Finance*, 137, 102915.